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Mark for Review



Levi and Marissa each have stamp collections. The number of stamps in Levi's collection is 250% of the number of stamps in Marissa's collection. If there are 300 stamps in Marissa's collection, how many stamps are in Levi's collection?

(A) 120

(B) 450

(C) 750

(D) 1,050





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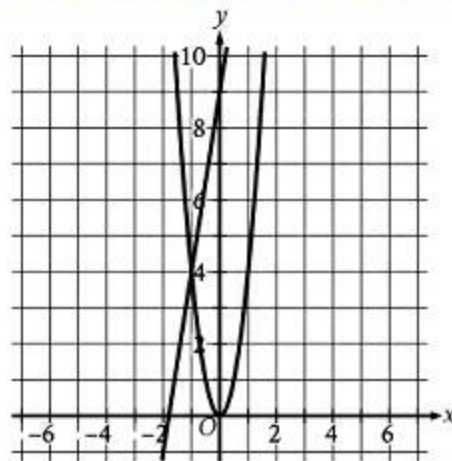


Reference



More

2.



The graph of a system of a linear equation and a quadratic equation is shown. Which of the following is a solution (x, y) to the system?

- A. $(-1, 4)$
- B. $(0, 0)$
- C. $(0, 9)$
- D. $(-1.8, 0)$

3



Mark for Review



An object moves at a speed of $\frac{3}{16}$ feet per second. What is this speed, in yards per second?
(3 feet = 1 yard)

(A) $\frac{1}{16}$

(B) $\frac{9}{16}$

(C) 6

(D) 16



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Mark for Review



Which expression is equivalent to $106t^3 - 26t^2u$?

(A) $2t(53t^2 - 13u)$

(B) $2t^2(53t - 13u)$

(C) $2tu(53t^2 - 13)$

(D) $2t^2u(53t - 13)$



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Mark for Review



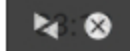
Circle K has a radius of 11 millimeters (mm). Circle L has an area of $144\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles K and L ?

(A) 23π

(B) 46π

(C) 92π

(D) 265π



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Mark for Review



Which expression is equivalent to $2\left(x - \frac{7}{2}\right)(x + 8)$?

(A) $2x^2 - 56$

(B) $2x^2 + 9x - 56$

(C) $2x^2 + x - 28$

(D) $x^2 + 9x - 28$





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Mark for Review



The function f is defined by $f(x) = (-2)(5)^x + 31$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

(A) $(0, 29)$



(B) $(0, 5)$

(C) $(0, -2)$

(D) $(0, 31)$



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Mark for Review



$$y < 24 - 6x$$

$$\frac{y}{6} > 9$$

Which inequality represents the x values for all solutions (x, y) that satisfy the given system of inequalities in the xy -plane?

(A) $x > 4$

(B) $x < 4$

(C) $x > -5$

(D) $x < -5$



9



Mark for Review



In the xy -plane, an equation of circle A is $(x - 4)^2 + (y - 8)^2 = 16$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A. Which equation represents circle B?

(A) $(x - 4)^2 + (y - 8)^2 = 32$

(B) $(x - 4)^2 + (y - 8)^2 = 64$

(C) $(x - 4)^2 + (y - 8)^2 = 128$

(D) $(x - 4)^2 + (y - 8)^2 = 256$



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Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

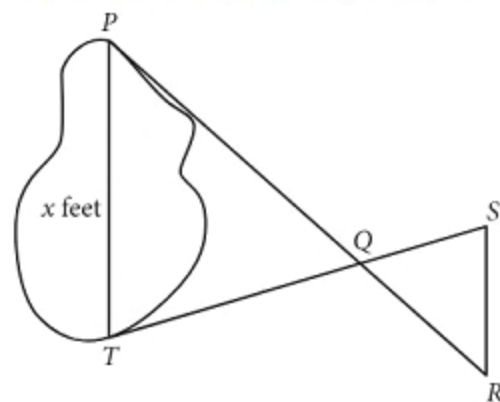
Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2

10



Mark for Review



Note: Figure not drawn to scale.

A property owner wants to find the length x , in feet, across a pond as represented in the figure. $\overline{PR}$ intersects $\overline{ST}$ at point Q , and $\angle PTQ$ is congruent to $\angle RSQ$. The lengths represented by PQ , TQ , QS , and RS were determined to be 4,900 feet, 9,800 feet, 1,400 feet, and 1,400 feet, respectively. What is the value of x ?

**11**

Mark for Review



In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{11}}{6}$. What is the value of $\cos(S)$?

(A) $\frac{\sqrt{11}}{6}$

(B) $\frac{\sqrt{11}}{5}$

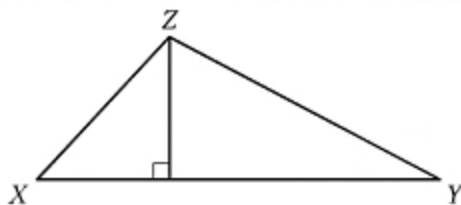
(C) $\frac{5\sqrt{11}}{11}$

(D) $\frac{6\sqrt{11}}{11}$



12

Mark for Review



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

(A) 247

(B) 494

(C) $247 \sin 54^\circ$

(D) $494 \sin 54^\circ$



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Mark for Review



A partially filled container containing 24 milliliters of water is placed under a leaky faucet that produces one 0.06-milliliter drop of water every 6 seconds. Until the container is full, which of the following can be used to represent the volume v , in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 6?

(A) $v = 6t$

(B) $v = 0.36t + 24$

(C) $v = 0.06t + 24$

(D) $v = 0.01t + 24$



14



Mark for Review



A research manager selected 2 random samples of ovens of a certain type to estimate the average amount of time this type of oven takes to preheat to 325 degrees Fahrenheit ($^{\circ}\text{F}$). The research manager recorded the amount of time, in minutes, each oven takes to preheat to 325 $^{\circ}\text{F}$. Based on the first sample, the research manager estimated that this type of oven takes an average of 14.2 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 1 minute. Based on the second sample, the research manager estimated that this type of oven takes an average of 14.4 minutes to preheat to 325 $^{\circ}\text{F}$, with an associated margin of error of 2.3 minutes. Assuming the margins of error were calculated the same way, which of the following best explains why the first sample obtained a smaller margin of error than the second sample?

- (A) The first sample contained fewer ovens than the second sample.
- (B) The first sample contained more ovens than the second sample.
- (C) The first sample took less time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.
- (D) The first sample took more time on average to preheat to 325 $^{\circ}\text{F}$ than the second sample.



Student-produced response directions

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Examples

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3.5	3.5 3.50 7/2	3 1/2 3 1/2

15



Mark for Review

$$-\frac{3}{14}rx + \frac{s}{9} = 10 - \frac{5}{56}x$$

In the given equation, r and s are constants. The equation has no solution. What is the value of r ?

Answer Preview:



Hide



Calculator



Reference



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Mark for Review



$$-5x(x + 5) = 15$$

One solution to the given equation can be written as $x = -\frac{s+\sqrt{t}}{2}$, where s and t are positive integers. What is the value of $\frac{s}{t}$?

(A) $\frac{5}{37}$

(B) $\frac{5}{22}$

(C) $\frac{5}{13}$

(D) $\frac{10}{13}$



17

Mark for Review



The equation $N(m) = 63(Q)^{\frac{m}{6}}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 180 seconds. What is the value of p in terms of Q ?

(A) $100(Q^{30} - 1)$

(B) $100(Q^{\frac{1}{2}} - 1)$

(C) $100(Q^{30} + 1)$

(D) $100(Q^{\frac{1}{2}} + 1)$



Student-produced response directions

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Examples

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3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2

18



Mark for Review

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,296 feet 9 seconds after it was launched. Based on the model, what was the height, in feet, of the object 3 seconds after it was launched?

Answer Preview:

Student-produced response directions

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Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ $3\frac{1}{2}$

19

 Mark for Review

Raheem bought 9 shirts that were each the same price. He used a coupon for \$54 off the entire purchase. The cost for the entire purchase after using the coupon was \$108. What was the original price, in dollars, for 1 shirt?

Answer Preview:



Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2

20



Mark for Review

The function p is defined by $p(x) = a((x+4)^2 - b)((x+4)^2 - c)$, where a , b , and c are constants. In the xy -plane, the graph of $y = p(x)$ passes through the points $(-5, 30)$ and $(0, 90)$. What is the value of $p(-8) + p(-3)$?

Answer Preview:



21



Mark for Review



$$27(x - n) = 27y + 27n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

(A) $2x - 2y = 4n$

(B) $2x + 2y = 2n$

(C) $2x + 2y = 4n$

(D) $2x - 2y = 2n$



Student-produced response directions

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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2

22



Mark for Review

$$(x - k)^2 = (k - 5a)(x - k)$$

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 32$. What is the value of a ?

I

Answer Preview: